MIL-STD-1312-17
26 JULY 1985
SUPERSEDING
MIL-STD-1312 (In part)
31 MAY 1967

MILITARY STANDARD

FASTENER TEST METHODS

METHOD 17,

STRESS RELAXATION

FSC 53GP

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE
WASHINGTON, DC 20301

Fastener Test Methods, Method 17, Stress Relaxation

MIL-STD-1312-17

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

FOREWORD

This standard sets forth a standard test method for stress relaxation testing threaded and non-threaded fasteners.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

TABLE OF CONTENTS

Paragraph		Page
1.	SCOPE	1
1.1	Applicability	1
2.	REFERENCED DOCUMENTS	1
2.1	Government documents	1
2.1.1	Specifications, standards and handbooks	1
2.2	Other publications	1
3.	DEFINITIONS	1
4.	GENERAL REQUIREMENTS	2
4.1	Test apparatus	2
4.1.1	Testing machines	2
4.1.1.1	Heating equipment and temperature control	2
4.1.1.2	Thermocouples	2
4.1.1.3	Temperature recording	2
4.1.1.4	Load measuring	2
4.1.1.5	Alignment	3
4.1.1.6	Loading rate	3
4.1.1.7	Load recording	3
4.1.1.8	Specimen holding fixtures	3
4.1.1.8.1	Fixture requirements	3
4.1.1.8.2	Fixture dimensions	3
5.	DETAIL REQUIREMENTS	3
5.1	Machine method test procedures	3
5.1.1	Load determination	3
5.1.1.1	Method	4
5.1.1.1.1	Threaded fasteners	4
5.1.1.1.2	Non-threaded fasteners	4
5.1.2	Relaxation machine set-up	5
5.1.2.1	Measured length of specimen	5
5.1.2.2	Relaxation determination	5
5.2	Cylinder method test procedures	5
5.2.1	Heating apparatus and temperature control	5
5.2.2	Test cylinders	5
5.2.2.1	Cylinder load determination	6
5.2.3	Elongation formula	6
5.2.4	Test fastener preparation	6
5.2.5	Measuring and loading procedures	7
5.2.6	Cylinder relaxation determination	7
6.	NOTES	8
6.1	Test report	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

TABLE OF CONTENTS (Continued)

FIGURES

Figure		Page
1.	Thermocouple assembly and location	10
2.	Plain test fixture and dimensions	11
3.	Threaded test fixture and dimension	12
4.	Torque versus induced load curve	13
5.	Load versus elongation curve	14
6.	Typical set-up for bolt relaxation only	15
7.	Typical set-up for nut and bolt relaxation	16
8.	Stress relaxation curve	17
9.	Load versus elongation curve	18

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Applicability. This test method outlines the procedure for stress relaxation testing threaded and non-threaded fasteners. The following methods designed to produce comparable results are specified:

a. Stress relaxation machine method, and

b. Stress relaxation cylinder method

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the current Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto (if applicable), form a part of this standard to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-45662 Calibration System Requirements

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 4 Load Verification of Testing Machines

ASTM E 83 Verification and Classification of Extensometers

ASTM E 220 Calibration of thermocouples by comparison techniques

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. DEFINITIONS

Not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

4. GENERAL REQUIREMENTS

4.1 Test apparatus.

4.1.1 Testing machines. The testing machines used shall be capable of maintaining the initial length of the loaded fastener within 0.0002-inch. This shall be accomplished by means of an extensometer which activates an unloading or loading mechanism whenever the initial stressed length changes by 0.0002-inch.

4.1.1.1 Heating equipment and temperature control. The heating equipment and temperature control shall be capable of maintaining the indicated temperature of the fastener over its entire length within $\pm 5^{\circ}F$ ($2.8^{\circ}C$) up to a test temperature of $2300^{\circ}F$ ($1260^{\circ}C$).

4.1.1.2 Thermocouples.

a. All thermocouples must be fabricated from calibrated thermocouple wire. The accuracy of this calibration shall be in accordance with ASTM E 220:

1. $\pm 2^{\circ}F$ ($1.1^{\circ}C$) from $0^{\circ}F$ ($-17.8^{\circ}C$) to $530^{\circ}F$ ($276^{\circ}C$).

2. $\pm 4^{\circ}F$ from $530^{\circ}F$ ($276^{\circ}C$) to $2300^{\circ}F$ ($1260^{\circ}C$).

b. Ceramic insulators or like materials shall be used from the thermocouple junction to at least 12 inches outside the furnace. The leads outside the furnace shall be covered with an insulating material.

c. The thermocouple shall be assembled so that the hot junction is held securely against the test fastener (see Figure 1) and the thermocouple performance not degraded by location.

d. A minimum of two thermocouples shall be attached to each test fastener. Where practical, one thermocouple shall be attached to each end, and one in the middle of the test fastener (see Figure 1).

e. Where radiation can affect the temperature reading, the thermocouple shall be protected by shielding.

4.1.1.3 Temperature recording. The temperature recording device shall be capable of maintaining a record of the temperature of the test fastener for the full duration of the test.

4.1.1.4 Load measuring. The load measuring device shall be capable of measuring the load applied to the test member with an accuracy of ± 2 percent for the full range of the machine capacity. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

calibration of the machine system shall conform to the requirements of MIL-STD-45662. Its accuracy shall be verified every 6 months by a method complying with ASTM E 4, using a calibration device calibrated by the National Bureau of Standards not more than 2 years before its use. Alternatively, the load measuring device may be calibrated in a universal testing machine whose calibration has been verified in accordance with ASTM E 4.

4.1.1.5 Alignment. The fixture and machines shall be capable of axial alignment. The alignment shall be checked with an alignment gage capable of measuring the elastic strain on four equally spaced places (located 90° apart, on a common axis) on the alignment gage surface. The axiality of alignment of machines and fixtures shall be such that an elastic strain difference of not more than 15 percent of the average strain shall be obtained on opposite sides of the alignment gage at the test load.

4.1.1.6 Loading rate. The axial load shall be applied to the test fastener at the rate of 100,000 pounds/minute ±10 percent per square inch of nominal shank area.

4.1.1.7 Load recording. The load recording device shall be capable of maintaining a continuous record of the residual load on the test fastener with an accuracy of ±2 percent for the full duration of the test. This instrumentation shall plot load versus time.

4.1.1.8 Specimen holding fixtures.

4.1.1.8.1 Fixture requirements. The stress relaxation test fixtures shall be capable of producing uniaxial loading. The hole shall be perpendicular to the bearing surface within 0.001-inch. The fixtures must be fabricated from materials capable of withstanding the test loads and temperatures without permanent deformation.

4.1.1.8.2 Fixture dimensions. The stress relaxation fixtures shall conform to those shown in Figure 2. When the fastener grip length is less than 2D, the fixtures in Figure 3 shall be used.

5. DETAIL REQUIREMENTS

5.1 Machine method test procedures.

5.1.1 Load determination. When the initial test load at test temperature is not provided in the product specification, or test authorization, it shall be determined from the following information, and utilized by the method described in 5.1.1.1.

a. A room temperature torque versus induced load curve similar to that shown in Figure 4 shall be generated for threaded fasteners. For non-threaded fasteners, the initial room temperature induced load shall be determined by an appropriate method.

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:39Z
Check the source to verify that this is the currentversion before use.

b. A room temperature load versus elongation curve similar to that shown in Figure 5 shall be generated for both threaded and non-threaded fasteners.

c. A load versus elongation curve shall be generated at the test temperature, or calculated from the modulus of elasticity of the test fastener material at the test temperature. These curves shall be plotted as shown in Figure 5.

5.1.1.1 Method. The method for determining the initial test load is as follows:

5.1.1.1.1 Threaded fasteners.

a. Determine the room temperature induced load (Load "A" in Figure 4) relative to the recommended tightening torque for the threaded fasteners. This load must be lower than the proportional limit "C" in Figure 4. The proportional limit is where the torque induced load curve deviates from a straight line.

b. If the recommended torque is not available, determine the load equal to 60 percent of the fastener minimum ultimate tensile strength. Develop a torque versus induced load curve (Figure 4) to determine the proportional limit "C." The initial test load must be lower than the proportional limit "C" in Figure 4.

c. Determine the elongation "X" (Figure 5) relative to the torque induced load "A" (Figure 4).

d. Determine the test temperature load "B" relative to the elongation "X" in Figure 5. The load "B" in Figure 5 is the recommended initial test load for relaxation tests conducted using the stress relaxation machine.

5.1.1.1.2 Non-threaded fasteners.

a. If the recommended load is not available, non-threaded fasteners shall be installed using their prescribed installation tools, and the resultant fastener load measured.

b. A room temperature load versus elongation curve similar to that shown in Figure 5 shall be generated.

c. Determine the elongation "X" Figure (5) relative to the fastener load determined in (a) above.

d. Determine the test temperature load "B" relative to the elongation "X" in Figure 5. The load "B" in Figure 5 is the initial test load for relaxation tests conducted using the stress relaxation machine.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

5.1.2 Relaxation machine set-up. Assemble the test member into the test fixture such that fixture faces do not touch each other. Attach extensometer and thermocouples to test member, in accordance with 5.1.2.1. Assemble the loaded fixture in the furnace, and seal furnace ends. After the test temperature has been reached, maintain temperature for 30 minutes before applying initial load. Apply initial load and adjust extensometer so that 0.0002-inch (0.0051 mm) increase in test member length will activate the unloading mechanism. The unloading should decrease the stress just enough to restore the test member to its original length, before the creep occurred. The test temperature and specimen length shall be maintained and load versus time recording shall be continued for the specified time. The temperature shall be monitored at sufficient intervals to assure conformance.

5.1.2.1 Measured length of specimen. The length measured by the extensometer in the testing machine shall be as follows:

a. The overall length of the bolt or pin, where the properties of only the bolt or pin are to be determined (Figure 6).

b. The length between bearing faces when the properties of the fastener system are to be determined (Figure 7).

5.1.2.2 Relaxation determination. The relaxation results may be reported as the final stress versus the initial stress for a given time and temperature if so specified in specification or test authorization. Unless so specified, the results shall be plotted as shown in Figure 8 where the residual stress (or load) is plotted against log time. When plotted as stress, it is recommended that the stress calculations be made using the pitch diameter or threaded members and the shank diameter after installation of non-threaded members. The area used shall be stated in the report.

5.2 Cylinder method test procedures. The cylinder method is only applicable to threaded fasteners.

5.2.1 Heating apparatus and temperature control. The heating apparatus and temperature control shall be capable of maintaining the indicated temperature of the heating zone of the furnace within $\pm 3^{\circ}F$ ($1.7^{\circ}C$) through $1200^{\circ}F$ ($649^{\circ}C$) and $\pm 5^{\circ}F$ ($2.8^{\circ}C$) over $1200^{\circ}F$ ($649^{\circ}C$).

5.2.2 Test cylinders. The cylinders shall be made from material having the same coefficient of thermal expansion and the same thermal treatment as the material in the test fasteners. The internal diameter of the cylinders should be 0.003-inch (0.08 mm) to 0.008-inch (0.20 mm) larger than the nominal test fastener diameter. The bearing surfaces shall be square to the I.D. within 0.003-inch FIM. The outside diameter of the cylinder shall be at least three times the basic major diameter of the test fastener.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

5.2.2.1 Cylinder load determination. Follow the method for determining the initial test load for threaded fasteners described in 5.1.1 and 5.1.1.1.1a, b, c and d. The elongation "X" and the load "B" in Figure 5 are the elevated temperature elongations and initial loads for relaxation tests conducted using the cylinder method.

5.2.3 Elongation formula. Elongation by calculation shall not be used unless it is known that the torque induced load at that elongation is lower than the proportional limit "C," in Figure 4.

5.2.3.1 The following formula may be used to determine the test elongation for threaded fasteners when the initial load at the test temperatures have been properly determined:

$$\Delta L = \frac{S}{E} \left[LC + \frac{3}{4} H + G \frac{(P.D.^2 - D^2)}{D^2} \right]$$

Where:

ΔL = Desired elongation (inches)

S = Desired stress at basic pitch diameter area of bolt thread in KSI

E = Modulus of elasticity of bolt material at room temperature

LC = Cylinder length (inches)

H = Maximum height of nut (inches)

P.D. = Basic pitch diameter of bolt thread (inches)

D = Nominal diameter of bolt shank (inches)

G = Nominal grip length of bolt (inches)

The room temperature and elevated temperature elongation shall be calculated and plotted similar to the load versus elongation curve in Figure 5.

5.2.4 Test fastener preparation. The ends of the fastener to be tested shall be prepared so that accurate, repeatable length measurements may be obtained.

a. The fastener ends may be ground parallel and marked to ensure proper orientation so that all measurements can be made at the same point; or the fastener ends can be spot drilled on centers at each end to provide an accurate contact with precision steel balls.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

b. The overall reproducibility of length measurements shall be maintained to within 0.0002-inch (0.005 mm).

5.2.5 Measuring and loading procedures.

1. Assemble the test fastener into the test cylinders.

2. Before applying any load, measure the length of the test fastener to the nearest 0.0001-inch (0.0025 mm). The length shall equal L_1.

3. Load the assembly to the desired elongation ±0.0001-inch (0.0025 mm) being careful to prevent overloading.

4. Allow the loaded assembly to remain at room temperature for one (1) hour.

5. Remeasure the loaded assembly and adjust as necessary to agree with the required elongation within 0.0001-inch.

6. This length shall be recorded as the loaded assembly length at room temperature before test. The length shall equal L_2.

7. The loaded assembly is then exposed to the test temperature for the required test time.

8. After removal from the heating apparatus, the assembly is allowed to cool to room temperature.

9. The loaded assembly is measured to the nearest 0.0001-inch (0.0025 mm) and this length is recorded as the loaded length after temperature exposure. The length shall equal L_3.

10. The assembly is unloaded and allowed to remain unloaded for a minimum of 30 minutes before measuring.

11. The unloaded assembly is measured to the nearest 0.0001-inch (0.0025 mm) and this measurement recorded as the unloaded length after temperature exposure. The measurement shall equal L_4.

5.2.6 Cylinder relaxation determination.

Joint relaxation: $L_5 = L_2 - L_3$

Bolt relaxation: $L_6 = L_4 - L_1$

Source: https://assist.dla.mil -- Downloaded: 202640-23T23:39Z
Check the source to verify that this is the currentversion before use.

$$\text{Total relaxation: } L_7 = L_5 + L_6 = (L_2 - L_3) + (L_4 - L_1)$$

$$\text{or } (L_2 - L_1) - (L_3 - L_4)$$

Where:

L_1 = The unstressed length of fastener, at room temperature, before testing.

L_2 = The stressed length of the assembly, at room temperature, before testing.

L_3 = The stressed length of the assembly, at room temperature, after testing.

L_4 = The unstressed length of the fastener, at room temperature, after testing.

L_5 = Relaxation of nut and/or cylinder (joint relaxation).

L_6 = Relaxation of bolt.

L_7 = Total relaxation.

5.2.6.1 Converting change in length into residual clamping load, or stress by reading the corresponding load or stress for the corresponding strain from Figure 5 (see example in Figure 9).

6. NOTES

6.1 Test report. The test report shall include the following data:

a. Fastener description.

1. Part number.
2. Lot identification.
3. Material.
4. Heat treat.
5. Grip length.
6. Mating part.
7. Measured fastener diameter.

b. Test machine.

1. Model and serial number.
2. Calibration date.

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:39Z
Check the source to verify that this is the current version before use.

c. Extensometer and gage length.

d. Installation procedure.

e. Test method (where options are available).

f. Test load.

g. Residual stress - final load or stress.

h. Test duration.

i. Type of failure.

j. Temperature reading.

k. Cause and length of interruption during test.

l. Results of all inspections.

m. Specimen preparation.

n. Measured length of cylinder, when used.

o. Stress versus time curve.

p. Torque versus load curve.

Custodians:
Army - AR
Navy - AS
Air Force - 11

Review activities:
Army - AV
Navy - AS, SH
Air Force - 99
DLA - IS

Preparing activity:
Navy - AS
(Project No. 53GP-0143-17)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

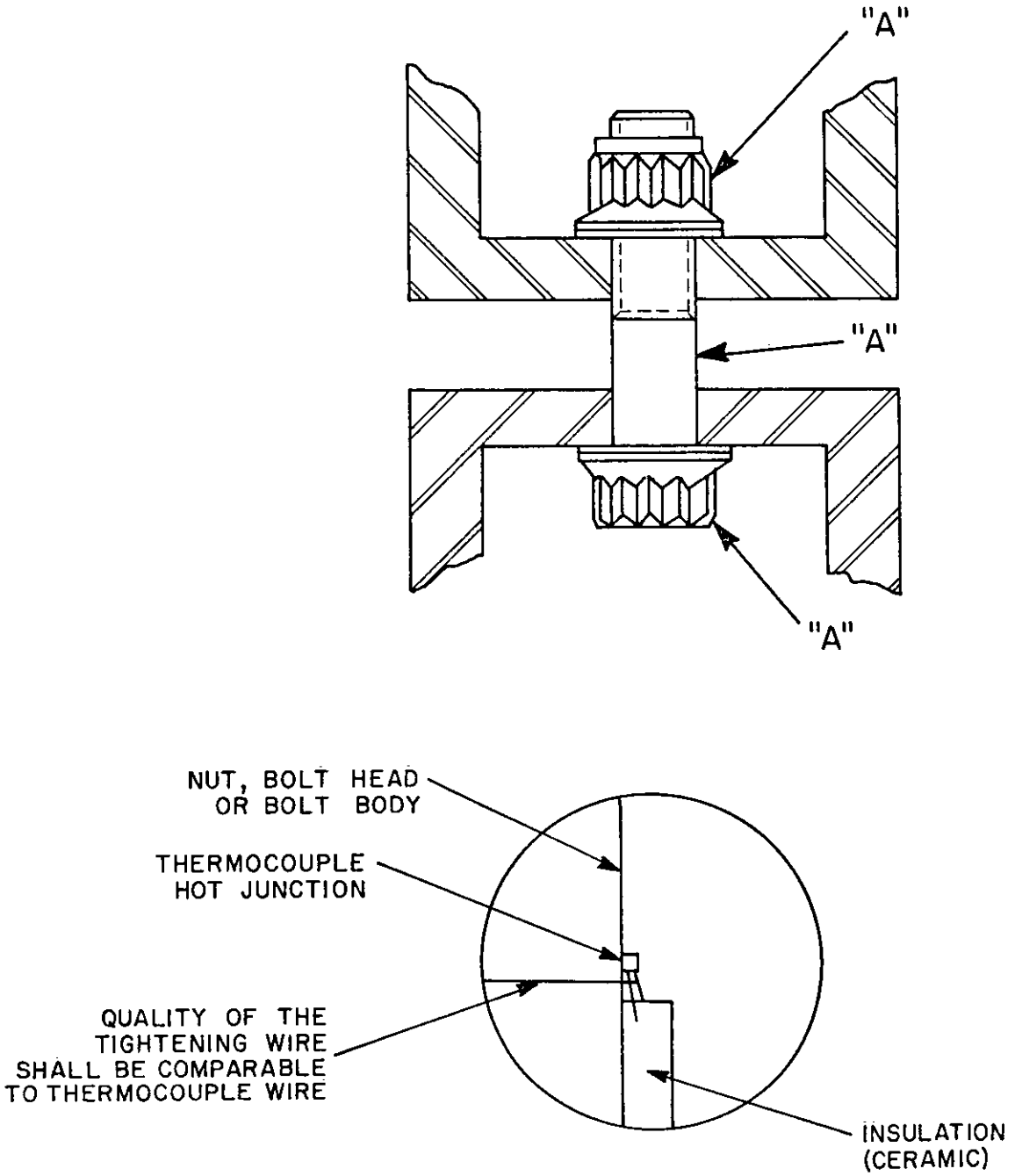
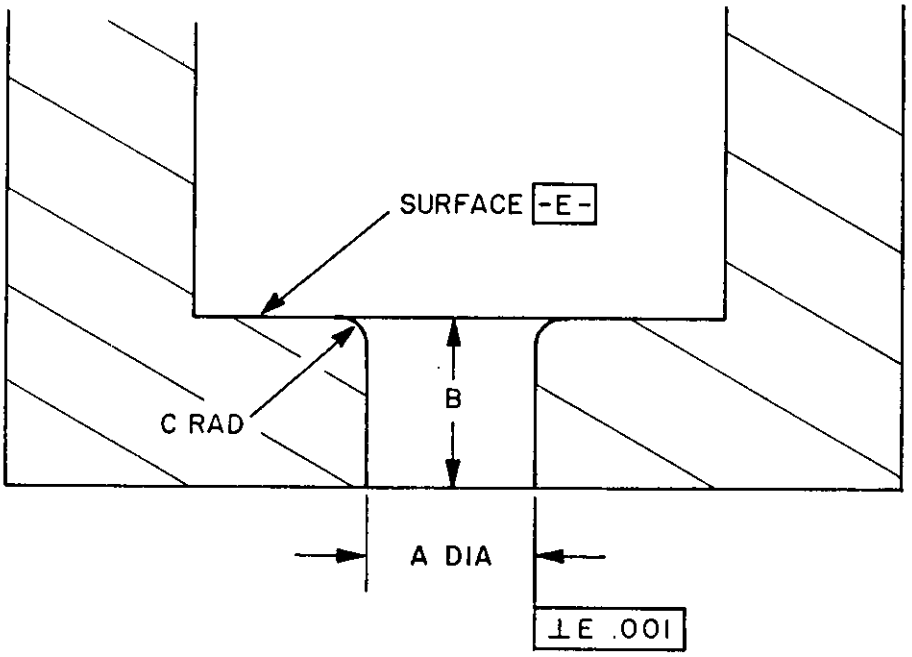


FIGURE 1. Thermocouple assembly and location.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

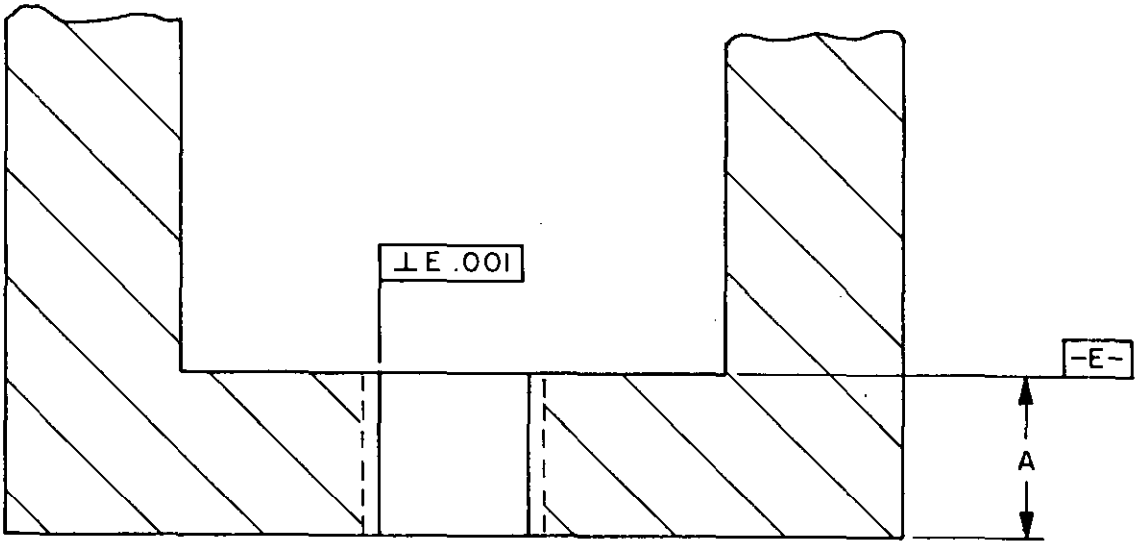


NOMINAL DIA.	A DIAMETER	B MIN	C RAD. ±0.005
TO 0.313	*D +0.005 -0.000	.8D*	0.050
0.375 - 0.563	D +0.010 -0.000	.8D	0.063
0.625 - 1.125	D +0.015 -0.000	.8D	0.080
1.250 - 1.500	D +0.020 -0.000	.8D	0.095

*D = NOMINAL DIAMETER

FIGURE 2. Plain test fixture and dimensions.

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:39Z
Check the source to verify that this is the current version before use.



SIZE	A MIN.	FINE THREAD		COARSE THREAD	
0.190	0.285	10-32	UNJF-3B	10-24	UNJC-3B
0.250	0.374	0.250-28	UNJF-3B	0.250-20	UNJC-3B
0.313	0.500	0.313-24	UNJF-3B	0.313-18	UNJC-3B
0.375	0.625	0.375-24	UNJF-3B	0.375-16	UNJC-3B
0.438	0.656	0.438-20	UNJF-3B	0.438-14	UNJC-3B
0.500	0.750	0.500-20	UNJF-3B	0.500-13	UNJC-3B
0.563	0.918	0.563-18	UNJF-3B	0.563-12	UNJC-3B
0.625	1.000	0.625-18	UNJF-3B	0.625-11	UNJC-3B
0.750	1.125	0.750-16	UNJF-3B	0.750-10	UNJC-3B
0.875	1.312	0.875-14	UNJF-3B	0.875-9	UNJC-3B

FIGURE 3. Threaded test fixture and dimension.

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:39Z
Check the source to verify that this is the current version before use.

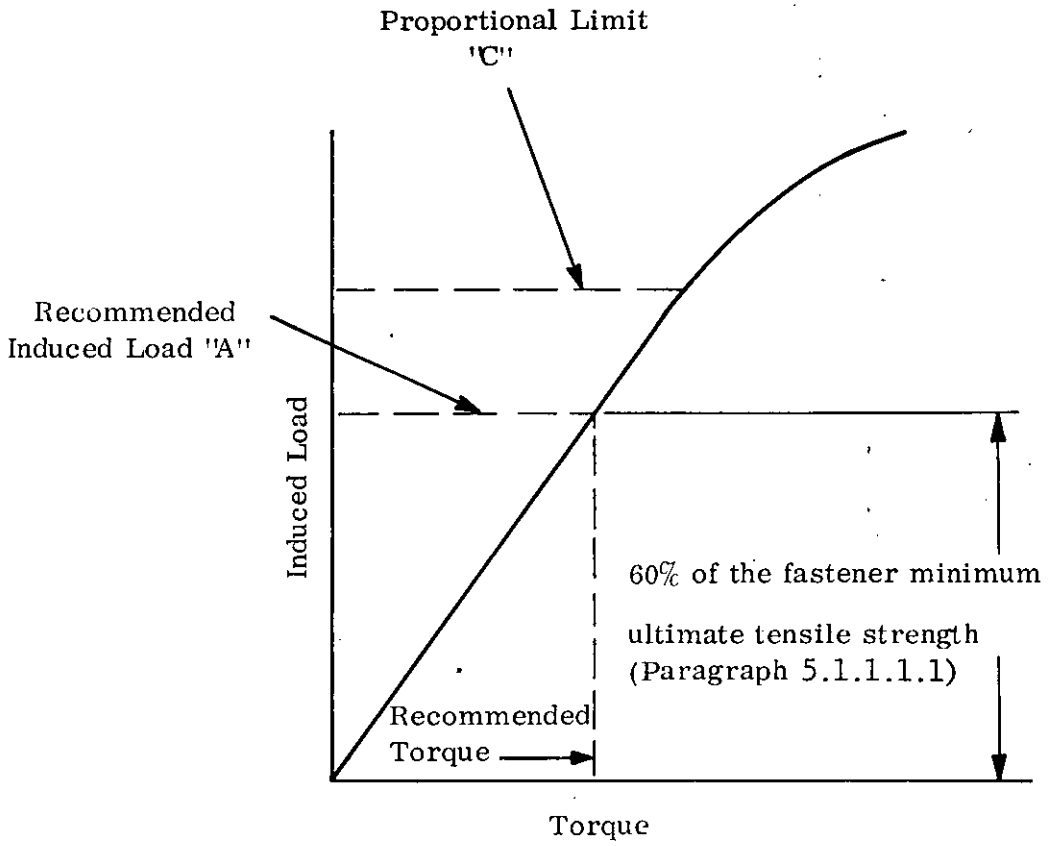


FIGURE 4. Torque versus induced load curve.

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:39Z
Check the source to verify that this is the current version before use.

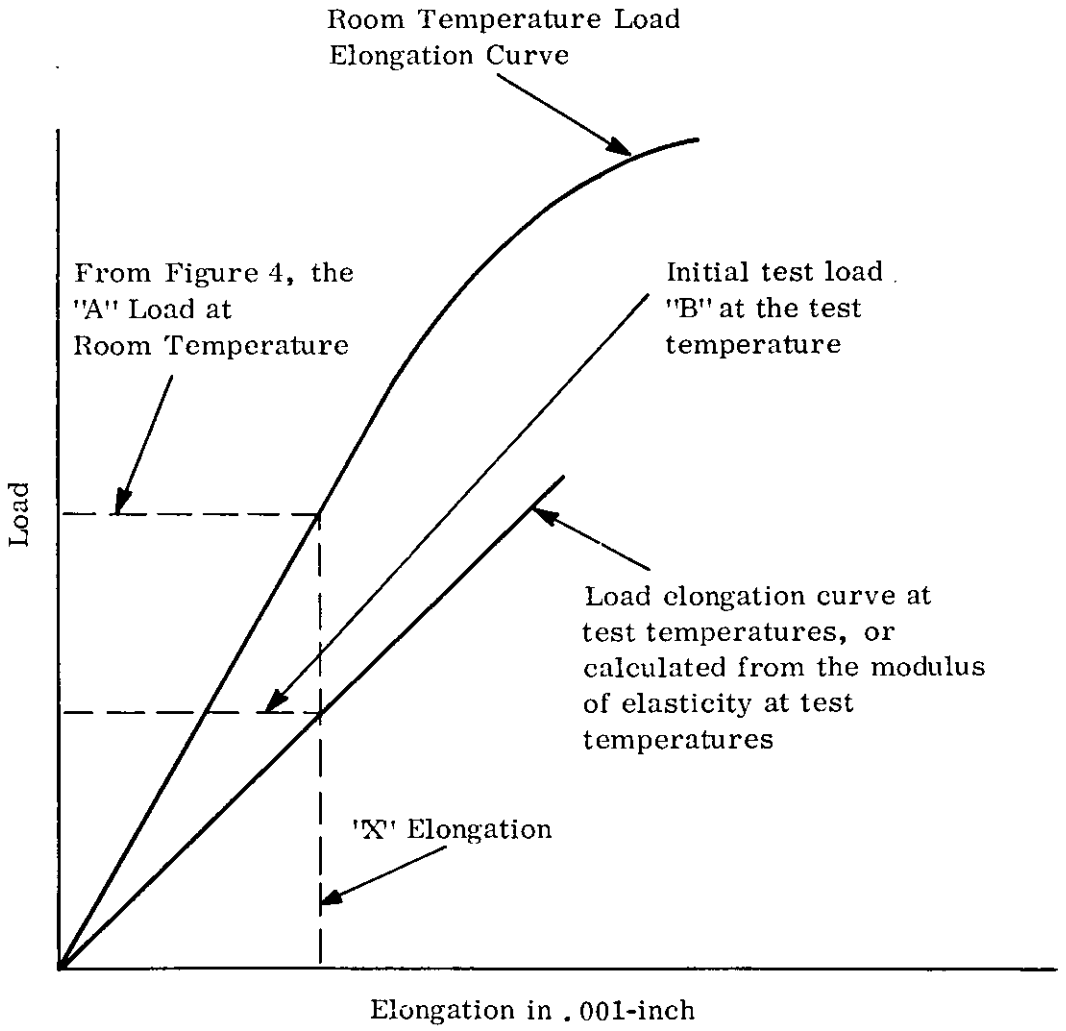


FIGURE 5. Load versus elongation curve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

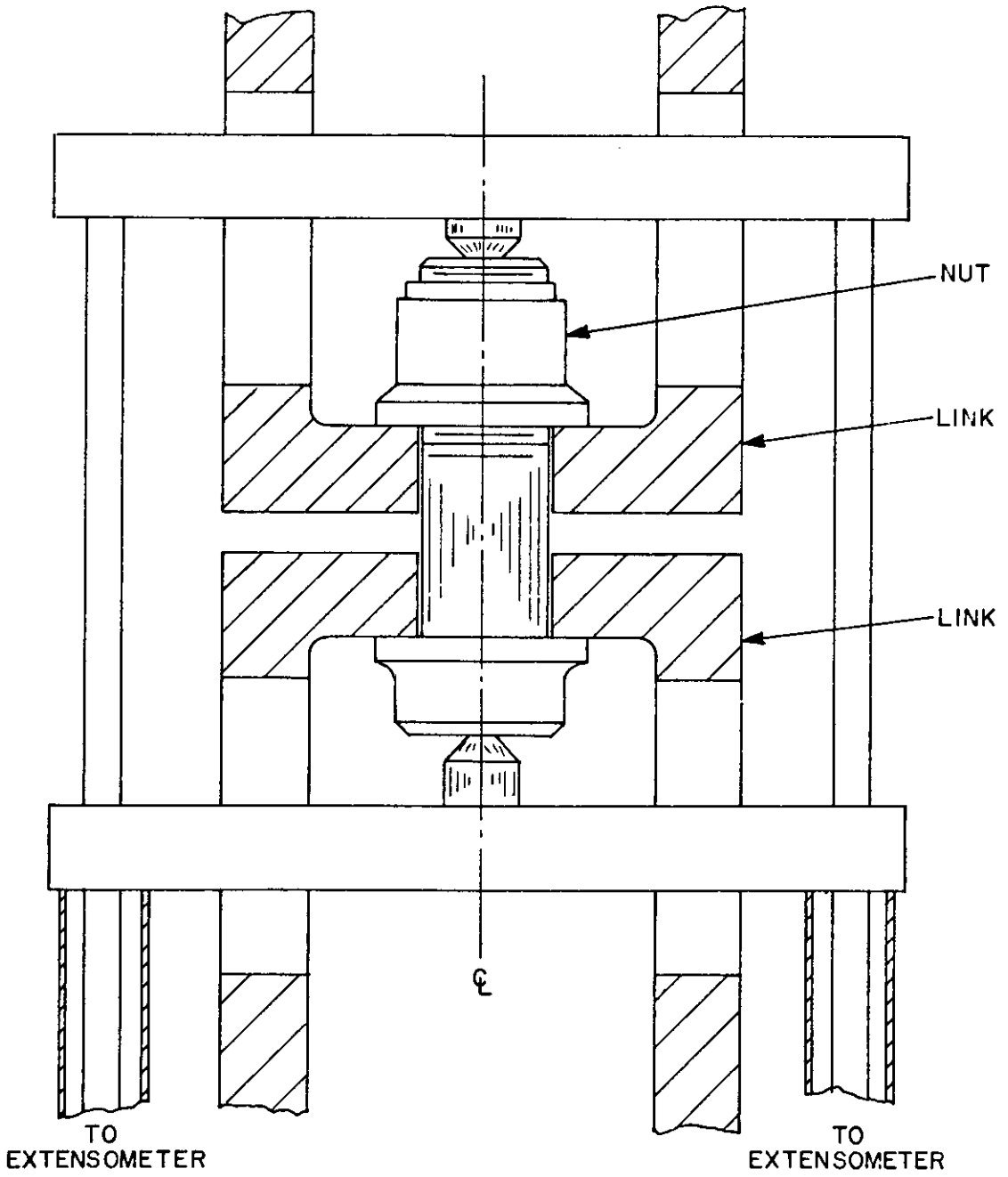


FIGURE 6. Typical set-up for bolt relaxation only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

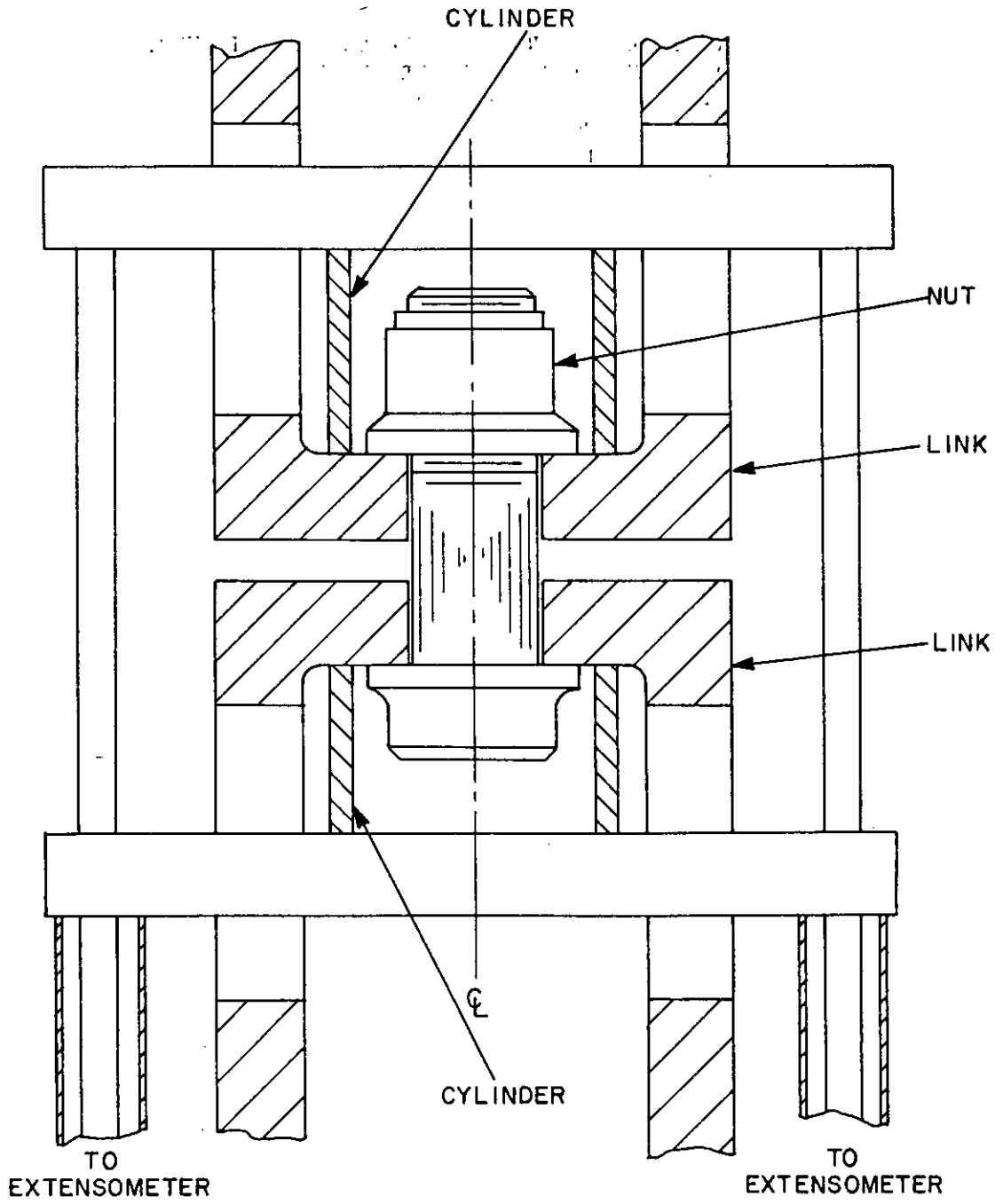


FIGURE 7. Typical set-up for nut and bolt relaxation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.

FASTENER DESCRIPTION INCLUDING PART NUMBER
TYPE OF NUT OR THREADED ADAPTER
MATERIAL AND HEAT TREAT
TEST TEMPERATURE
TYPE MACHINE OR CYLINDERS

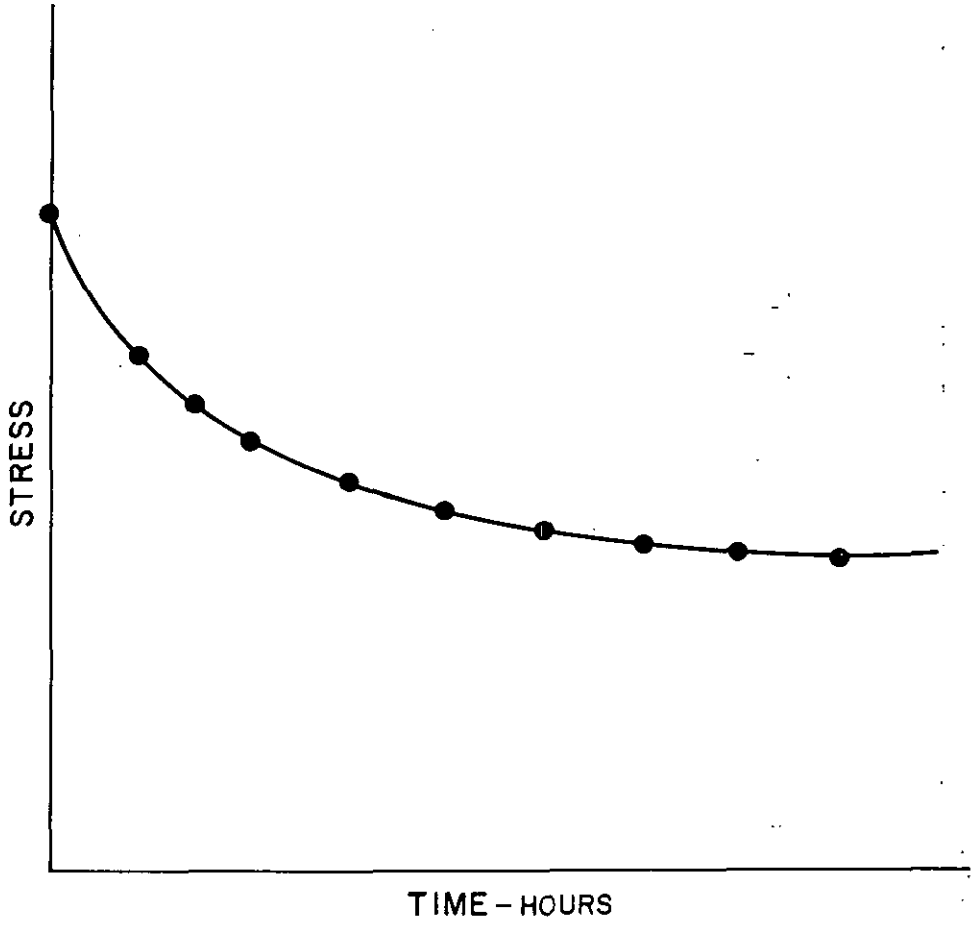
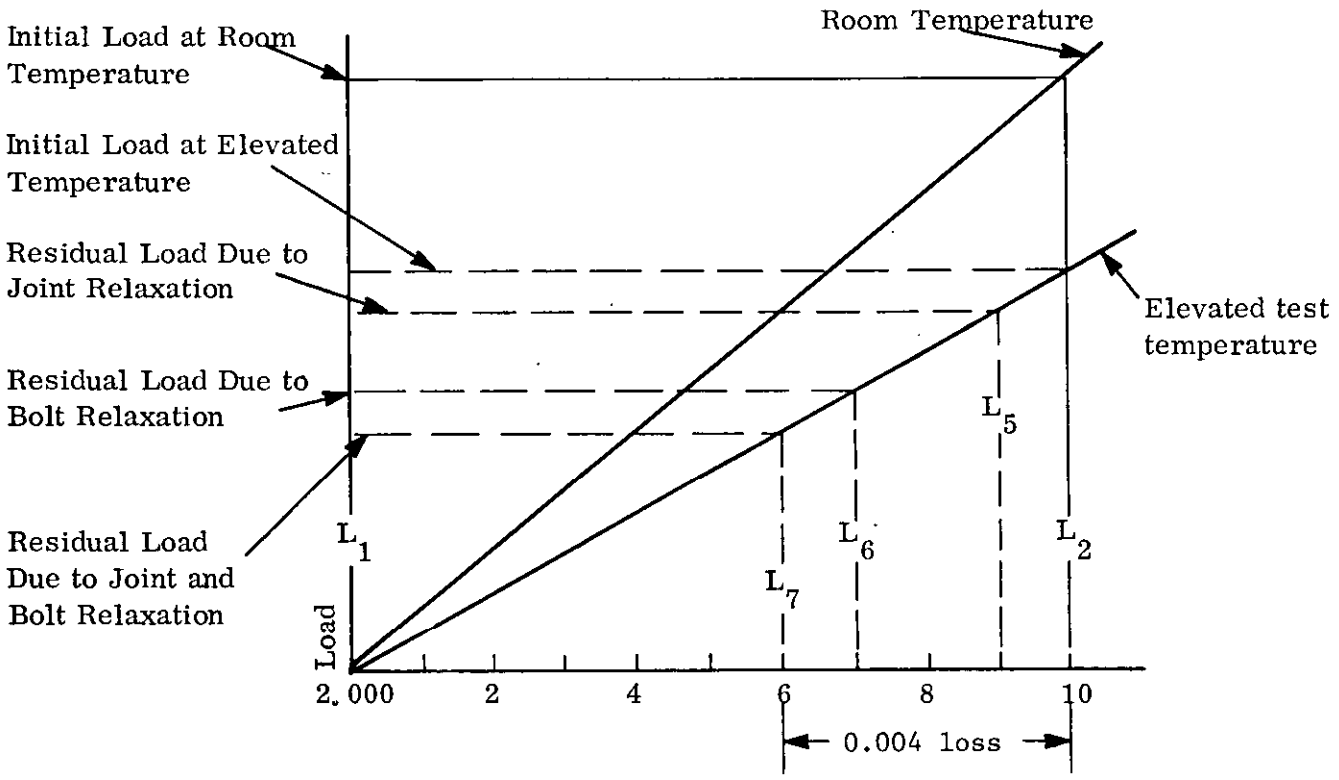


FIGURE 8. Stress relaxation curve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.



Example: All dimensions in inches

	L_1	L_2	L_3	L_4
Fastener Length -	2.000	2.010	2.009	2.003

Joint Relaxation $L_5 = L_2 - L_3 = 2.010 - 2.009$

L_5 = 0.001 elastic strain loss resulting in residual load at 2.009 inches elongation.

Bolt Relaxation $L_6 = L_4 - L_1 = 2.003 - 2.000$

L_6 = 0.003 elastic strain loss resulting in residual load at 2.007 inches elongation.

Total Relaxation $L_7 = L_5 + L_6 = 0.001 + 0.003$

L_7 = 0.004 elastic strain loss resulting in residual load at 2.006 inches elongation.

The residual loads in this example are at the elevated test temperature. The residual loads at room temperature can easily be determined by projecting L_5, L_6, and L_7 up to the room temperature load vs. elongation curve.

FIGURE 9. Load versus elongation curve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:39Z
Check the source to verify that this is the current version before use.